

Technology Stack

Date	3 July 2026
Team ID	xxxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Technology Stack Table

Layer	Technology Chosen	Version	Justification
Frontend	HTML5, Bootstrap 5, JavaScript	Bootstrap 5.3.2	Responsive, mobile-first design with no heavy frontend framework overhead. Bootstrap provides a professional banking UI out of the box.
Backend	Python, Flask	Python 3.12, Flask 3.1.2	Flask is a lightweight, flexible web framework ideal for ML model serving. Python is the standard language for ML pipelines.
ML Engine	Scikit-learn, XGBoost, Joblib	sklearn 1.8.0, XGBoost 3.2.0	Industry-standard ML libraries. Scikit-learn provides LR, DT, RF classifiers; XGBoost adds gradient boosting. Joblib handles fast model serialization.
Data Processing	Pandas, NumPy, Seaborn,	Pandas 2.3.3, NumPy 2.3.4	Pandas handles large CSV ingestion and preprocessing. NumPy for numerical operations. Seaborn/Matplotlib for EDA

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	Matplotlib		visualizations.
Cloud / Hosting	Render	Free tier	Zero-cost Flask deployment with automatic GitHub integration. Supports Python runtime with Gunicorn WSGI server. Singapore region for low latency.
Version Control	Git + GitHub	—	Team collaboration, commit history, and CI/CD pipeline. Auto-deploy to Render on every push to master branch.
Dataset Source	Kaggle	rikdifos/credit-card-approval-prediction	Publicly available, CC0 licensed dataset with 438K+ real applicant records and 1M+ monthly credit records.
Development Tools	Jupyter Notebook, VS Code	—	Jupyter for EDA and model experimentation. VS Code for application development.